

Overview of network attacks and potential detection implementations:

Name: Distributed Denial of Service (DDOS)

Description: Overwhelm a system with lots of fraudulent traffic from various locations.

Traffic: Lots of traffic at once coming from several compromised locations

Detection: Create a decision boundary for when traffic is too much so likely is a DDOS attack. In a DOS attack it will all come from the same place so a smaller boundary might be used to detect faster. In a DDOS attack, traffic from lots of unrecognised places might start appearing all at once. It might be faster to detect a lot of random sources appearing or a larger amount of traffic from several distributed sources.

Name: Man in the middle (MITM)

Description: Attackers act as a relay for the data and can view and manipulate it in real time.

Traffic: Little difference to the traffic at the other end as source IP is unchanged.

Detection: Possible detection through consistent increase of latency indicating the traffic is taking a longer route or size of data is different to what is expected.

Name: IP Spoofing

Description: An attacker fakes the IP address of packets to that of an allowed IP address, therefore gaining access into the network.

Traffic: The traffic will look like it is coming from a valid source so very hard to identify from the packets.

Detection: Might be detectable through time of access. If there is access at the same time as similar IP addresses it might be strange as they would not need to access the system for very separate tasks simultaneously. Another potential giveaway might be the packets arrive a lot faster or slower than would be expected of that IP address.

Name: Insider attack

Description: Someone (or some malware) inside the network has access to the data and sends it out of the network to a secondary location where they can use the files maliciously.

Traffic: Packets with a source within the network will have a destination address not recognised.

Detection: Can be detected by checking destination address against accepted addresses. In systems where this isn't possible it might be possible to check for large data dumps going out of the network or a steady data flow over a longer period.